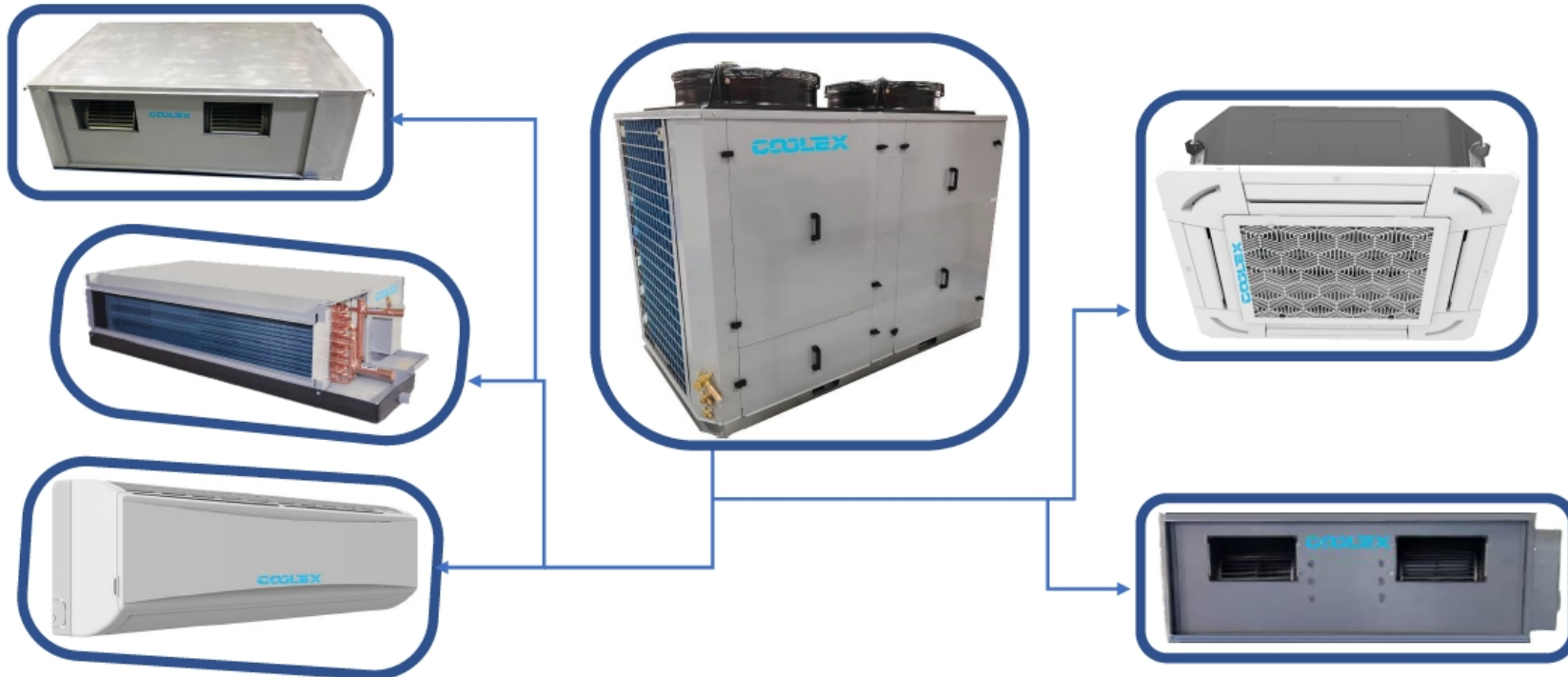


VRF SYSTEM DESIGN REPORT AND PERFORMANCE SHEET



Content

Project Information..... 3

Air-conditioning design parameters..... 3

Quotation list..... 4

Air-conditioning system..... 6

 System 1..... 6

 System 2..... 10

Project Information

Project Information	
Project Name	Kuwait City, Sharq
Project Location	KuwaitCity
Construction Area (m²)	0
Customer	Company
Customer Telephone	9999999
Customer Email	@ric.com.kw
Project Consultant	Emad
Project Designer	Fahad

Air-conditioning design parameters

Air-conditioning design parameters			
Summer	Summer atmospheric pressure	101,325	Pa
	Summer outdoor DB	47	°C
	Summer indoor DB	27	°C
	Summer indoor WB	13	°C
Winter	Winter atmospheric pressure	101,325	Pa
	Winter outdoor DB	8	°C
	Winter Outdoor WB	4.94	°C
	Winter indoor DB	20	°C
Altitude		801	m

Quotation list

Equipment Quotation

Model	Type	Quantity	Unit	Price ()	Subtotal ()
ODU					
IVCF-00956CDH	IVCF SERIES	1	pc	0	0
IVCF-01546CDH	IVCF SERIES	1	pc	0	0
IDU					
ICEF-00224DH	4-Way Cassette	1	pc	0	0
ICEF-00364DH	4-Way Cassette	1	pc	0	0
ICEF-00484DH	4-Way Cassette	2	pc	0	0
IVHF-00164DH	Concealed Ducted High Static	1	pc	0	0
IVHF-00244DH	Concealed Ducted High Static	1	pc	0	0
IVHF-00304DH	Concealed Ducted High Static	1	pc	0	0
IVLF-00304DH	Concealed Ducted Low Static	1	pc	0	0
Branch joint					
RN-01	Branch joint	1	Piece	0	0
RN-02	Branch joint	4	Piece	0	0
RN-03	Branch joint	1	Piece	0	0
Wired controller					
Wired	Wired controller	4	pc	0	0
Remote controller					
Remote	Remote controller	4	pc	0	0
	Total	24			0

Installation Material Quotation

Model	Quantity	Unit	Price ()	Subtotal ()
Total	0			0

Total quotation price

Air-conditioning Quotation	Price ()
Equipment	0
Installation materials	0
Total	0

Air-conditioning system

System 1

System Information

System Information			
Total room numbers	3	Air conditioning area (m²)	0
ODU model	IVCF-00956CDH	IDU quantity	3
Nominal cooling capacity @35°C (W)	28,293.08	Nominal heating capacity @7°C (W)	28,293.08
Rated cooling capacity @47°C (W)	19,756	Rated heating capacity @8°C (W)	26,799
Nominal Cooling combination ratio @35°C (%)	105%	Rated cooling @47°C (%)	100%
Nominal heating combination ratio @7°C (%)	86%	Rated heating combination ratio @8°C (%)	88%
Nominal system cooling EER @35°C (W/W)/(Btu/W.h)	4.09/13.96	Nominal system heating COP @7°C (W/W)	4.78
Rated system cooling EER @47°C (W/W)/(Btu/W.h)	2.85/9.74	Rated system heating COP @8°C (W/W)	4.41
*Additional refrigerant charge (kg)	0	Own refrigerant (kg)	22

*The above data is for reference only, the actual additional refrigeration weight depends on the actual pipe length.

Length/Height difference limit

Item	Current value	Limit value
Total pipe length (m)	0	1,000
Length between ODU and the farthest IDU (m)	0	200
The pipe length ODU to the outside first branch joint (m)	0	10
Length between ODU's 1st branch joint and farthest IDU (m)	0	90
Length difference between farthest IDU and nearest IDU (m)	0	40
IDU-IDU height difference (m)	0	30
ODU-IDU height difference (m)	23	110
IDU-ODU height difference (m)	0	130

Indoor Unit List

Floor	Room	Room area (m²)	Room cooling/heating load (W)	IDU Model	Quantity	Nominal cooling capacity @35°C (W)	Nominal heating capacity @7°C (W)	Rated cooling capacity @47°C (W)	Rated heating capacity @8°C (W)	Rated air flow rate (m³/h)	Rated ESP (Pa)	Control	Pump	Panel
Floor 2	Room 4	0	5000/0	IVHF-00244DH	1	7,000	8,300	4,679.4	6,368.92	1,700	25	Wired	N	
Floor 2	Room 5	0	0/0	IVLF-00304DH	1	8,500	9,200	5,800.89	8,631.63	1,615	10	Wired	N	
Floor 2	Room 3	0	0/0	ICEF-00484DH	1	14,100	15,200	9,275.24	11,798.25	2,175	0	Remote	N	

Outdoor Unit Parameters

ODU model	Nominal cooling capacity @35°C (W)	Nominal heating capacity @7°C (W)	Rated cooling capacity @47°C (W)	Rated heating capacity @8°C (W)	Nominal cooling / heating power input (W)	Rated cooling / heating power input (W)	Quantity	Nominal EER @35°C /COP @7°C (W/W)	Rated EER @47°C /COP @8°C (W/W)	gas/ liquid (mm)	Own refrigerant (kg)	Power	PF	MCA	MFA	Weight (kg)	Noise (dB(A))	Outer dimension (HxWxD) (mm)
IVCF-00956CDH																		
IVCF-00956CDH	28,293	28,293	19,756	26,799	6,630/5,636	6,637/5,794	1	4.27/5.02	2.98/3.41	22.2/12.7	22	380~415V , 3P	0.95	36.6	65	280	58	1,643x1,126x1,096

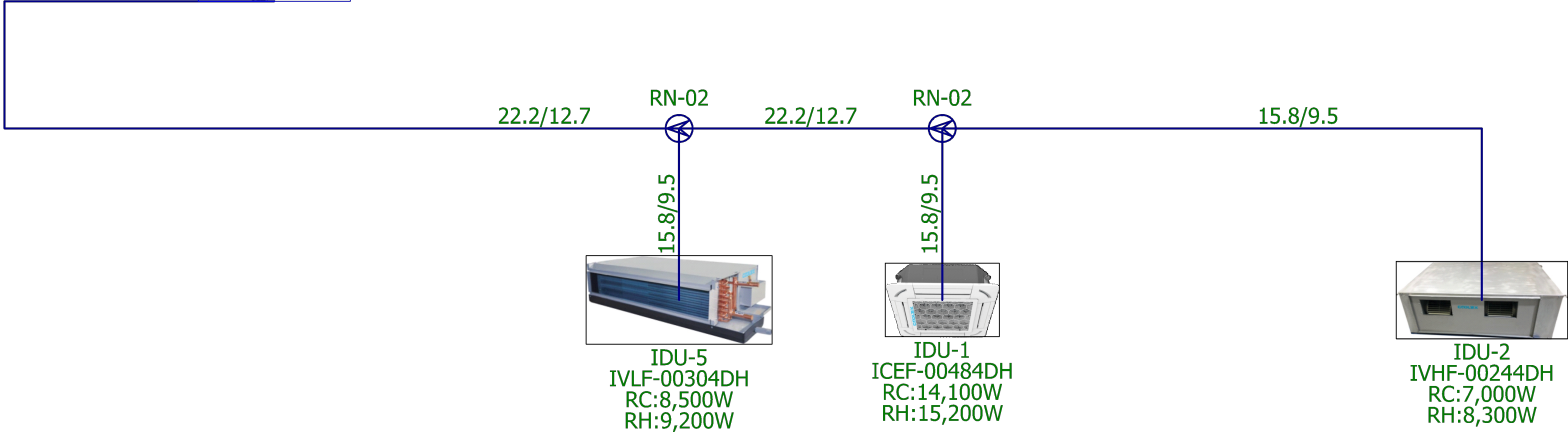
Indoor Unit Parameters

IDU Model	Type	Nominal cooling / heating capacity (W)	Nominal power input (W)	Air volume (m³/h)	Static pressure (Pa)	Power	MCA (A)	Weight (kg)	Noise (dB(A))	Outer dimension(HxWxD) (mm)
ICEF-00484DH		14,100/15,200	95	2,175/1,560/945	0	220~240V , 1P	5.88	5.5	46	290x845x845
IVHF-00244DH		7,000/8,300	80	1,700/1,593/1,486	25	220~240V , 1P	3.13	63	45	356x1,270x855

IVLF-00304DH		8,500/9,200	110	1,615/760/710	10	220~240V , 1P	3.13	33	30	250x1,377x550
--------------	--	-------------	-----	---------------	----	---------------	------	----	----	---------------

Refrigerant drawing

System 1
IVCF-00956CDH
RC:28,293W RH:28,293W



System 2

System Information

System Information			
Total room numbers	5	Air conditioning area (m²)	0
ODU model	IVCF-01546CDH	IDU quantity	5
Nominal cooling capacity @35°C (W)	44,707.99	Nominal heating capacity @7°C (W)	44,707.99
Rated cooling capacity @47°C (W)	31,910	Rated heating capacity @8°C (W)	44,898
Nominal Cooling combination ratio @35°C (%)	100%	Rated cooling @47°C (%)	100%
Nominal heating combination ratio @7°C (%)	89%	Rated heating combination ratio @8°C (%)	101%
Nominal system cooling EER @35°C (W/W)/(Btu/W.h)	3.33/11.37	Nominal system heating COP @7°C (W/W)	3.9
Rated system cooling EER @47°C (W/W)/(Btu/W.h)	2.68/9.15	Rated system heating COP @8°C (W/W)	3.97
*Additional refrigerant charge (kg)	0	Own refrigerant (kg)	30

*The above data is for reference only, the actual additional refrigeration weight depends on the actual pipe length.

Length/Height difference limit

Item	Current value	Limit value
Total pipe length (m)	0	1,000
Length between ODU and the farthest IDU (m)	0	200
The pipe length ODU to the outside first branch joint (m)	0	10
Length between ODU's 1st branch joint and farthest IDU (m)	0	90
Length difference between farthest IDU and nearest IDU (m)	0	40
IDU-IDU height difference (m)	6	30
ODU-IDU height difference (m)	0	110
IDU-ODU height difference (m)	9	130

Indoor Unit List

Floor	Room	Room area (m²)	Room cooling/heating load (W)	IDU Model	Quantity	Nominal cooling capacity @35°C (W)	Nominal heating capacity @7°C (W)	Rated cooling capacity @47°C (W)	Rated heating capacity @8°C (W)	Rated air flow rate (m³/h)	Rated ESP (Pa)	Control	Pump	Panel
Floor 2	Room 5	0	0/0	ICEF-00364DH	1	10,600	11,400	7,275.23	9,642.65	2,175	0	Remote	N	
Floor 3	Room 1	0	0/0	IVHF-00164DH	1	4,800	5,300	3,643.86	5,043	1,190	25	Wired	N	
Floor 3	Room 2	0	0/0	IVHF-00304DH	1	8,800	10,200	6,449.81	9,116.49	2,040	25	Wired	N	
Floor 4	Room 7	0	0/0	ICEF-00224DH	1	6,400	7,600	4,731.79	6,887.44	1,275	0	Remote	N	
Floor 4	Room 8	0	0/0	ICEF-00484DH	1	14,100	15,200	9,809.76	13,344.66	2,175	0	Remote	N	

Outdoor Unit Parameters

ODU model	Nominal cooling capacity @35°C (W)	Nominal heating capacity @7°C (W)	Rated cooling capacity @47°C (W)	Rated heating capacity @8°C (W)	Nominal cooling / heating power input (W)	Rated cooling / heating power input (W)	Quantity	Nominal EER @35°C /COP @7°C (W/W)	Rated EER @47°C /COP @8°C (W/W)	gas/ liquid (mm)	Own refrigerant (kg)	Power	PF	MCA	MFA	Weight (kg)	Noise (dB(A))	Outer dimension (HxWxD) (mm)
IVCF-01546CDH																		
IVCF-01546CDH	44,708	44,708	31,910	44,898	12,897/10,962	11,389/10,781	1	3.47/4.08	2.8/2.96	28.6/15.8	30	380~415V, 3P	0.88	66.7	0	504	64	1,635x2,105x1,160

Indoor Unit Parameters

IDU Model	Type	Nominal cooling / heating capacity (W)	Nominal power input (W)	Air volume (m³/h)	Static pressure (Pa)	Power	MCA (A)	Weight (kg)	Noise (dB(A))	Outer dimension(HxWxD) (mm)
ICEF-00224DH		6,400/7,600	60	1,275/660/45	0	220~240V, 1P	3.13	5.5	43	245x845x845
ICEF-00364DH		10,600/11,400	95	2,175/1,560/945	0	220~240V, 1P	3.13	5.5	45	290x845x845

ICEF-00484DH		14,100/15,200	95	2,175/1,560/945	0	220~240V , 1P	5.88	5.5	46	290x845x845
IVHF-00164DH		4,800/5,300	80	1,190/992/793	25	220~240V , 1P	3.1	40	44	334x1,167x636
IVHF-00304DH		8,800/10,200	185	2,040/2,039/1,835	37	220~240V , 1P	3.13	63	46	356x1,270x855

Refrigerant drawing

